Slip 02

Q1) A Differentiate between virus, worm, and Trojan horse. .

1. Introduction

A virus, worm, and Trojan horse are three types of malware (malicious software) that can harm computers, damage data, steal information, or affect system performance. However, they differ in their working methods, spreading mechanisms, and effects.

2. Virus

A computer virus is a malicious program that attaches itself to a legitimate file or program and spreads when the infected file is executed or shared.

Features:

- Requires a host file or program.
- Can replicate by infecting other files.
- May corrupt or delete files.
- Usually requires some user action to spread.

Example: A file-infecting virus that infects executable files on a computer.

3. Worm

A computer worm is a standalone malicious program that can replicate itself and spread automatically from one computer to another, often through a network.

Features:

- Does not require a host file.
- Can spread automatically across networks.
- May consume network bandwidth and system resources.
- Can spread rapidly and affect many computers.

Example: The Conficker worm, which spread across Windows computers by exploiting security vulnerabilities.

4. Trojan Horse

A Trojan horse is a type of malware that disguises itself as legitimate or useful software to trick users into installing or running it.

Features:

- Appears to be genuine or useful software.
- Tricks users into executing it.
- May steal sensitive information or provide unauthorized access.
- Does not usually replicate itself automatically.

Example: A fake antivirus application that appears to protect a computer but secretly installs malicious software.

5. Differences Between Virus, Worm, and Trojan Horse

Basis	Virus	Worm	Trojan Horse
1. Definition	Malware that attaches to a file or program.	Malware that replicates and spreads independently.	Malware disguised as legitimate software.
2. Host requirement	Requires a host file or program.	Does not require a host file.	Usually exists as a standalone program.
3. Replication	Replicates by infecting other files.	Replicates automatically.	Does not usually self-replicate.
4. Spreading method	Through infected files, USB drives, or downloads.	Through networks and vulnerable systems.	Through deceptive downloads, attachments, or applications.
5. User interaction	Often requires execution of an infected file.	Can spread without direct user action.	Usually requires the user to install or run it.
6. Main purpose	Infect files and potentially damage data.	Spread rapidly and disrupt systems or networks.	Trick users and perform malicious activities.

7. Speed of spreading	Depends on file sharing and execution.	Can spread rapidly across networks.	Depends mainly on user installation or execution.
8. System impact	May corrupt, modify, or delete files.	May consume bandwidth and system resources.	May steal data or enable unauthorized access.
9. Detection	Detected through antivirus scanning and file analysis.	Detected through antivirus and network monitoring.	Detected through antivirus and suspicious application analysis.
10. Example	File-infecting virus	Conficker worm	Fake antivirus containing malware

6. Prevention Methods

1. Install and regularly update antivirus software.
2. Keep the operating system and applications updated.
3. Avoid opening suspicious email attachments or links.
4. Download software only from trusted sources.
5. Use a firewall and perform regular data backups.

7. Conclusion

Viruses, worms, and Trojan horses are dangerous forms of malware. A virus infects files, a worm spreads independently, and a Trojan horse deceives users by pretending to be legitimate software. Understanding these differences helps users choose appropriate cybersecurity measures and protect their systems from attacks.

Q. 2 B

Write a Python program to detect spam emails using predefined keywords.

```
spam_keywords = [
"win",
"winner",
"free",
"prize",
"lottery",
"urgent",
"offer",
"click here",
"claim",
"money",
"congratulations"
]
email = input("Enter the email message: ")
email_lower = email.lower()
spam_words_found = []
for keyword in spam_keywords:
if keyword in email_lower:
spam_words_found.append(keyword)
print("\n---------- Result ----------")
if len(spam_words_found) >= 2:
print("Email Status: SPAM")
print("Spam keywords found:", spam_words_found)
else:
print("Email Status: NOT SPAM")
print("Spam keywords found:", spam_words_found)
print("----------------------------")

#Output
#Enter the email message: Congratulations! You won a free prize.
Click
here to claim your money.
#---------- Result ----------
#Email Status: SPAM
#Spam keywords found: ['free', 'prize', 'click here', 'claim',
'money',
'congratulations']
#----------------------------
```

OR

Q2)
B
Write a Python program to encrypt user input text using AES encryption

```
from Crypto.Cipher import AES
from Crypto.Util.Padding import pad

# Accept plaintext from user
plaintext = input("Enter the plaintext: ")

# AES key - 16 bytes (128-bit)
key = b"1234567890123456"

# Create AES cipher object
cipher = AES.new(key, AES.MODE_ECB)

# Convert plaintext into bytes and add padding
data = pad(plaintext.encode(), AES.block_size)

# Encrypt the plaintext
encrypted_text = cipher.encrypt(data)

# Display encrypted text in hexadecimal form
print("Original Text :", plaintext)
print("Encrypted Text:", encrypted_text.hex())
```